

How can we formulate our research questions effectively?

Yongjin Park

13 January 2026

**Please fill out the survey
of your background**

- ASDA
- <https://asda.stat.ubc.ca/2025/11/10/free-stats-help.html>
- If you and your team are interested in the following ASDA projects, let me know.
- I will send you one-page project proposal document.

ASDA #1: Blood RNAs for Febrile Illness Diagnosis

- Scoping review identified 286 RNA signatures from literature
- Signatures diagnose or predict outcomes of febrile illnesses (sepsis, TB, infections)
- Goal: Characterize and understand potential sources of biases
- Meta-analysis across many different studies

ASDA #2: Breath print data for Lung Cancer Detection

- Breath samples from 500+ participants (lung cancer vs. healthy controls)
- Volatile organic compounds (VOCs) analyzed via mass spectrometry
- High-dimensional data: ~500-1500 compounds per sample
- Goal: Build analysis pipeline for biomarker discovery and classification

Roadmap of STAT/BIOF/GSAT540 2026

data generation followed by *inference* or *validation*

- ② *Statistics and matrix algebra*
- ③ **Formulating biological questions** (Today)
- ④ **Hypothesis testing** (Thursday)
- ⑤ RNA-seq data analysis
- ⑥ Statistical inference
- ⑧ Single-cell genomics
- ⑨ Matrix factorization
- ⑩ Genome-wide association studies
- ⑪ Multiple hypothesis testing

A biological problem $\Rightarrow$
a computational problem

Guide 1: Crystallize abstract ideas into a concrete computational problem by writing a paper

Research Model #1

- 1 Idea
- 2 Do research
- 3 Write a paper

Research Model #2

- Idea
- **Write a paper**
- Do research

Which model would work better for you? Why?

Guide 1: Crystallize abstract ideas into a concrete computational problem by writing a paper

Research Model #1

- 1 Idea
- 2 Do research
- 3 Write a paper

Research Model #2

- Idea
- **Write a paper**
- Do research

Which model would work better for you? Why?

Discussion with your peers around you (5 min).

Guide 1: Crystallize abstract ideas into a concrete computational problem by writing a paper

Research Model #1

- 1 Idea
- 2 Do research
- 3 Write a paper

Research Model #2

- Idea
- **Write a paper**
- Do research

Which model would work better for you? Why?

- 1 Model #2 forces us to be clear, focused
- 2 Crystallizes what we don't understand yet (and what we don't need to know)
- 3 Opens the way to dialogue with others: reality check and collaboration

Simon Peyton Jones, How to write a great research paper

Model #1 or #2?

Guide 2: Ten simple rules ...

William Noble,
**Ten simple rules for defining a
computational biology project.**
PLoS Comput Biol (2023)

Summary: Ten Simple Rules

- 1 Write it down
- 2 Define the problem
- 3 Define potential impact
- 4 Summarize related work
- 5 State central hypothesis
- 6 Sketch your approach
- 7 Identify available data
- 8 Choose validation measures
- 9 Set up version control
- 10 Share with colleagues

Discussion: How many items have you followed in the past?

Rule 1: Write it down

Document your idea in “writing”

- Clarifies your thinking and the underlying problem
 - Exposes half-baked concepts early
 - Forces you to articulate assumptions
- “The act of writing your idea down will inevitably clarify it for you.”*

Even a rough draft is better than no draft

Rule 2: Define the problem

Clearly delineate the scope and nature of the problem

- What exactly are you trying to solve?
- What is **in scope** vs. **out of scope**?
- Be specific, not vague
 - ma

Which sounds more specific?

- Understand gene expression programs in myeloid cancer
- Predict enhancer-promoter interaction in K562 cells

Rule 4: Summarize related work

Conduct literature review early

“A terrible trap to fall into is carrying out an entire research project only to find that the same work was already published before you began.”

- Learn from others' approaches
- Identify gaps in current knowledge
- **Summarize** methods in a chronologically ordered table

Rule 5: State central hypothesis

Reduce your idea to a single sentence

- Focus on the one, single, **overarching** theme
- Make it **testable**
- Keep it **simple and clear**
- It takes practice and time.

Rule 7: Identify available data

Ensure you have both inputs and validation data

- What are the input?
- How do we know that our discovery is correct?
- Simulation studies are okay.

Rule 9: Set up version control

Use `git` or a similar tool from day one

- Track code evolution from project inception
 - Keep the dates
- Enable collaboration
- Maintain reproducibility
- Document your progress

Rule 10: Share with colleagues

Seek feedback on feasibility and scientific importance

- Prepare to “Think Again” (don’t be embarrassed or offended by being wrong)
- Refine your ideas through discussion
- Find potential collaborators

Ten Simple Rules (items) → A paper

- ① **Write it down**
 - ② **Define the problem**
 - ③ Define potential impact
 - ④ Summarize related work
 - ⑤ **State central hypothesis**
 - ⑥ Sketch your approach
 - ⑦ **Identify available data**
 - ⑧ Choose validation measures
 - ⑨ Set up version control
 - ⑩ Share with colleagues
- Abstract
 - Rule #1, #2, #5
 - Introduction
 - Rule #2, #3, #4, #5
 - Results
 - Rule #5, #7, #10
 - Discussion
 - Rule #10
 - Methods
 - Rule #6, #7, #8, #9, #10

It's so critical to describe your problem clearly.

It's so critical to describe your problem clearly.

- Recognize the underlying computational problem(s)

It's so critical to describe your problem clearly.

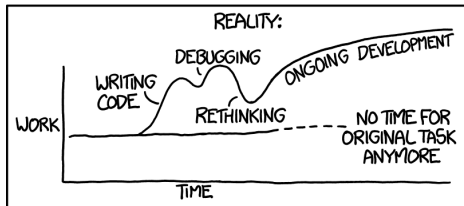
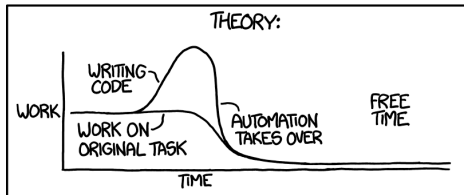
- Recognize the underlying computational problem(s)
- Make an actionable plan to solve them.

It's so critical to describe your problem clearly.

- Recognize the underlying computational problem(s)
- Make an actionable plan to solve them.
- Evaluate what/how it worked and didn't.

Is defining a problem worth my time?

"I SPEND A LOT OF TIME ON THIS TASK.
I SHOULD WRITE A PROGRAM AUTOMATING IT!"



Two flavors

- #1: Optimization approach
 - **The art of abstraction**
 - Define **input**, **output**, and **metric** (objective function)
 - Find the same or similar existing problems in the literature
- #2: Probabilistic Modelling
 - **Think of a potential generative model**
 - Describe the problem in terms of random variables
 - Structural Equation Model (in epidemiology)
 - Use an existing inference algorithm

Today's lecture

- 1 Define a problem as an optimization problem
- 2 Use probabilities to describe data generation process

The art of abstraction: Same data, different questions

Single-cell RNA-seq data: a matrix of **genes** $\times$ **cells**

Biological questions $\rightarrow$ computational problems

Exercise: Eleven Grand Challenges in Single-Cell

Lähnemann et al., *Genome Biology* (2020)

Transcriptomics

- ➊ Handling sparsity (dropout)
- ➋ **Defining cell identity**
- ➌ Tracking cellular dynamics
- ➍ Mapping to reference atlases

Genomics

- ➎ Finding variants (scDNA-seq)
- ➏ Identifying copy-number changes
- ➐ Inferring regulatory networks

Phylogenetics

- ➑ Understanding tumor evolution
- ➒ Calling structural variants

Integration

- ➓ Multi-modal integration
- ➔ Scaling to millions of cells

Let's work on one of these examples

Example: Defining cell identity (Challenge 2)

Biological question: What cell types are in my sample?

- What are the variables of interest?
- Is it important?

How should we think of cells in this context?

Data: $m \times n$ expression matrix **X** (m genes $\times$ n cells)

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$$

where m corresponds to 19k genes.

How should we think of cells in this context?

Data: $m \times n$ expression matrix **X** (m genes $\times$ n cells)

- Is a cell a vector of gene expressions over 19k genes?
- Is a cell a vector of gene expressions over several genes?
- Is a cell a point in a "cell type landscape?"
- ...

Let's start writing out the problem

Biological question: What cell types are in my sample?

Input

Gene expression X (m genes $\times$ n cells)

Output

Assign each cell vector $\mathbf{x}_j$ to label $c_i \in \{1, \dots, K\}$

Brainstorming ...

- How did the cell types emerge in multicellular organisms?
- How did the cell express genes?
- What do we measure with a cell-level gene expression matrix?
- How do we know cells belong to a cell type A, not B?

Example: Let's write it down with several sentences

- ① We can consider each cell as a point along a differentiation trajectory.
- ② We can represent the relationships between cells as a graph (connecting points).
- ③ Cells that share similar expression profiles are closely located in a graph space.

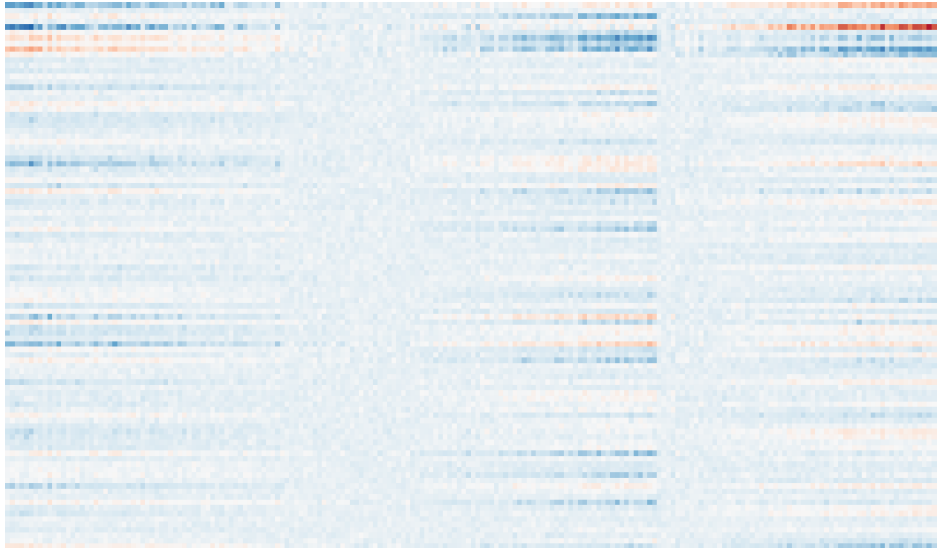
Can we do better? Is it specific enough?

How do we measure the distance between cells?

Multiple levels of distance measures between two cell vectors:

- ① **Use all genes:** Compute distance using all m features (e.g., Euclidean, cosine)
- ② **Focus on selected genes:** Use only marker genes or highly variable genes
- ③ **Learn new coordinates:** Project cells into a lower-dimensional space (e.g., PCA) and compute distance there

A toy example: 100 genes and 200 cells/samples

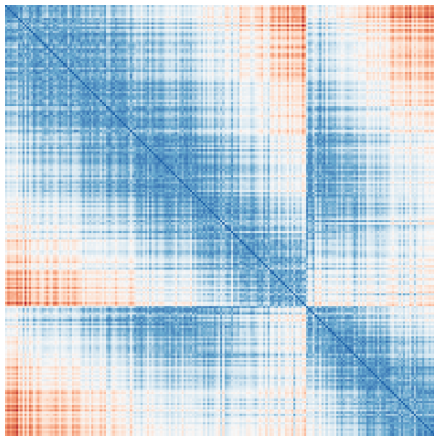


Compute cell-cell distance in Top 5 PCA space

- row = cells;
column =
cells

Note: *Don't
worry about what
PCA is yet
(lect-09)*

Instead of dealing
with 100 (genes)
dimensions, we
only need to deal
with 5
dimensions.



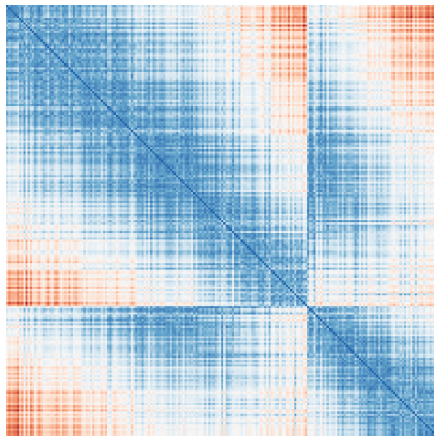
blue (small); red (large)

Compute cell-cell distance in Top 5 PCA space

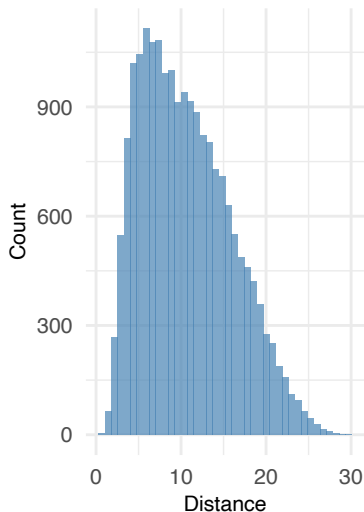
- row = cells;
column =
cells

Note: *Don't worry about what PCA is yet (lect-09)*

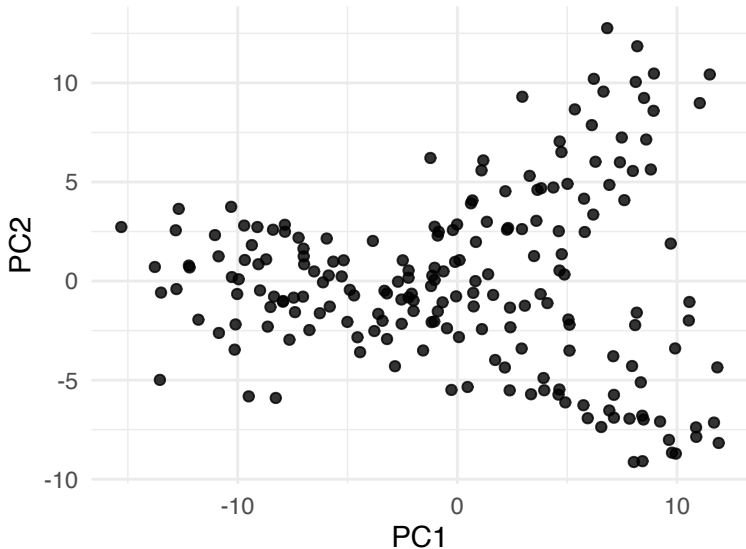
Instead of dealing with 100 (genes) dimensions, we only need to deal with 5 dimensions.



blue (small); red (large)



Visualize cells in PC space

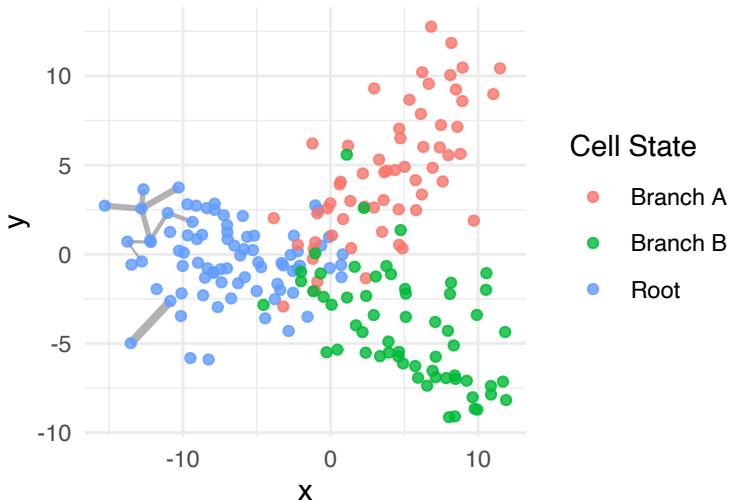


Reminder: Our goal is to build a graph of cells

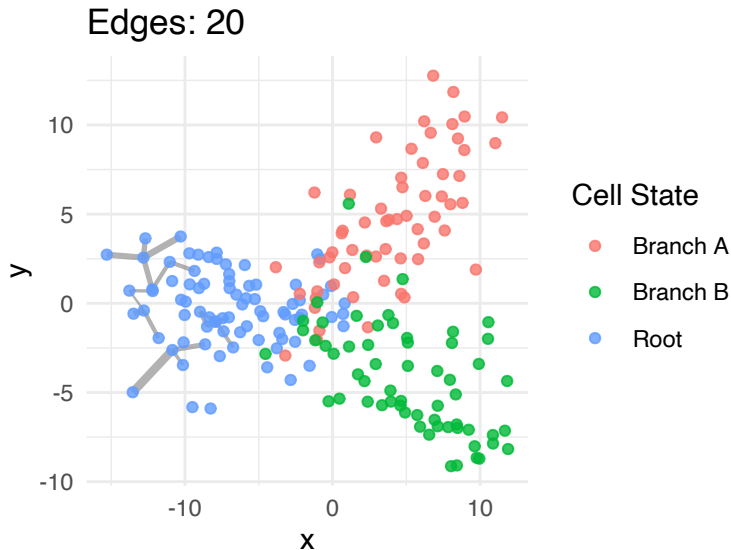
- Graph = connecting dots (dot = cell)
- Tree = a special case of graph
- Minimum Spanning Tree = one way to connect the dots
- What is our assumption?

Building Minimum Spanning Tree

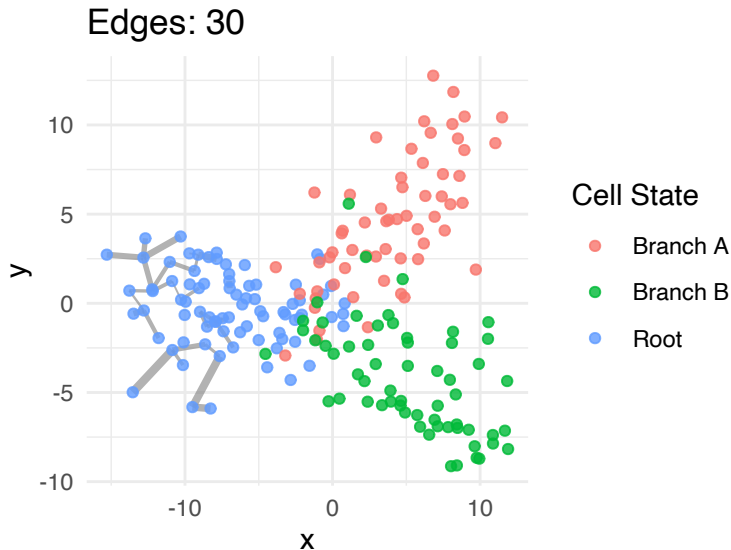
Edges: 10



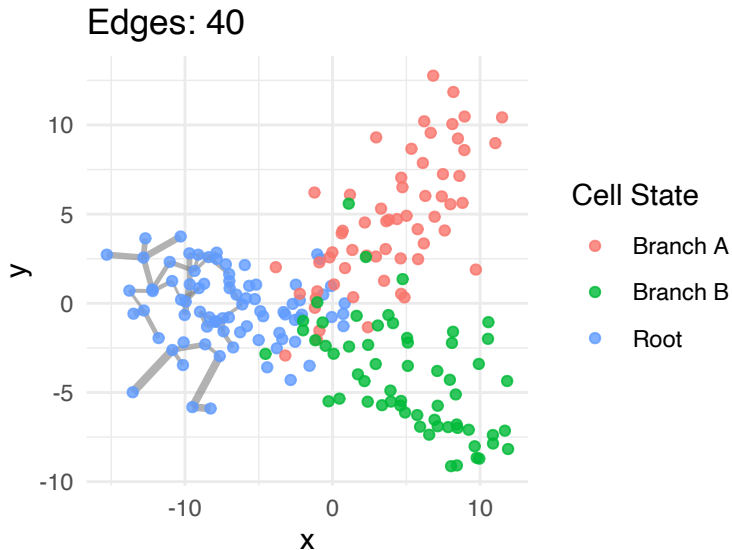
Building Minimum Spanning Tree



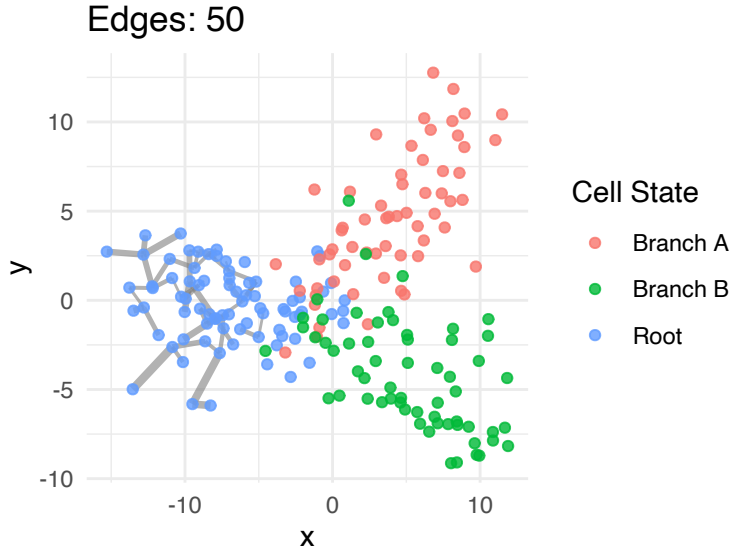
Building Minimum Spanning Tree



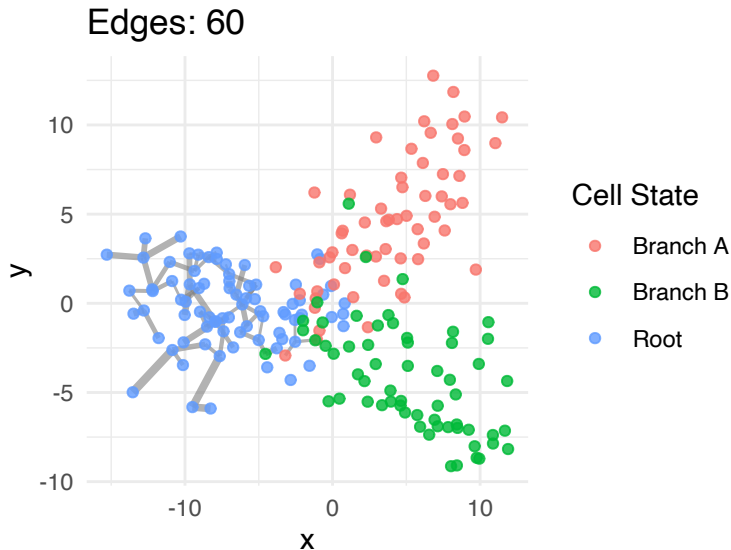
Building Minimum Spanning Tree



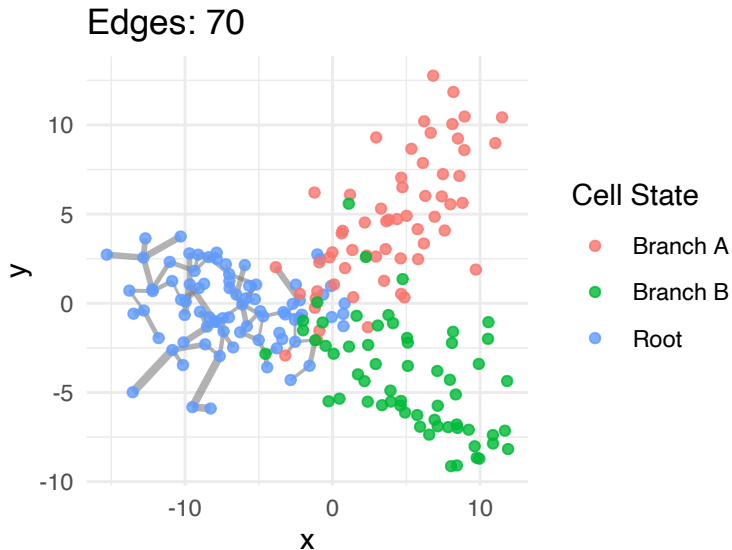
Building Minimum Spanning Tree



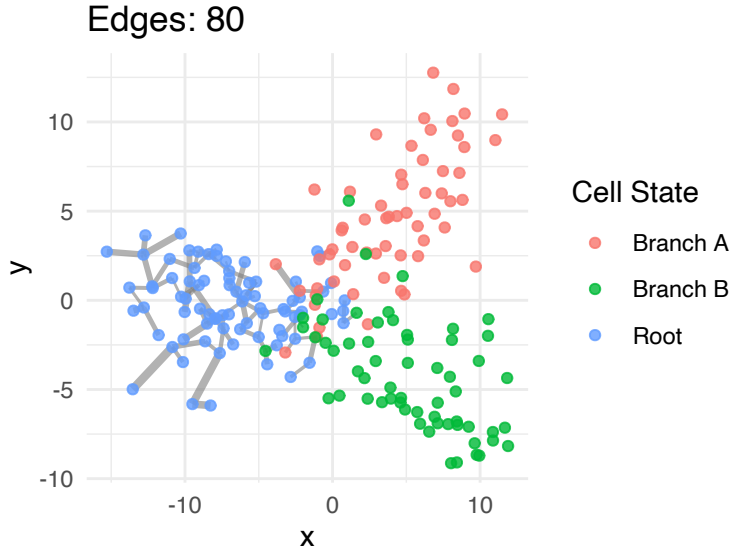
Building Minimum Spanning Tree



Building Minimum Spanning Tree

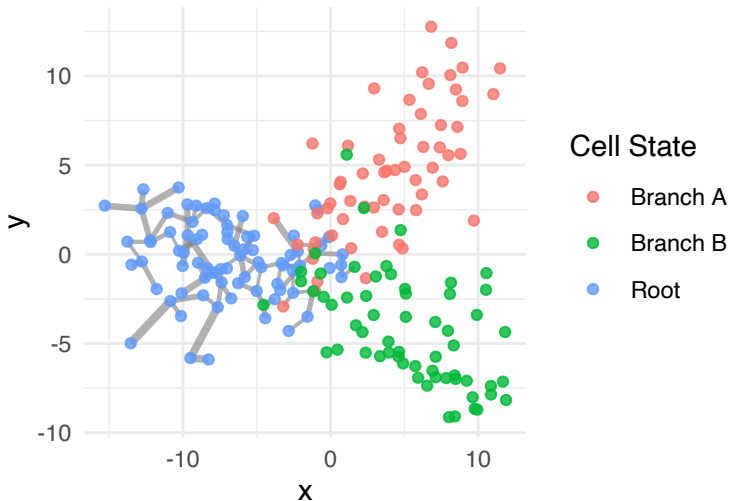


Building Minimum Spanning Tree



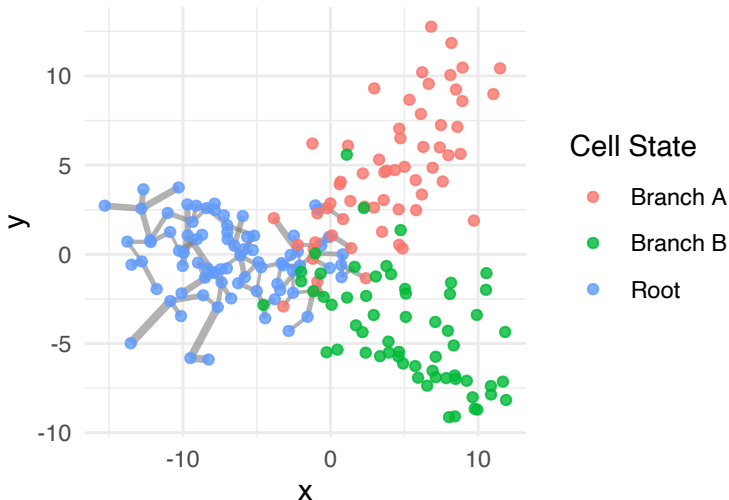
Building Minimum Spanning Tree

Edges: 90

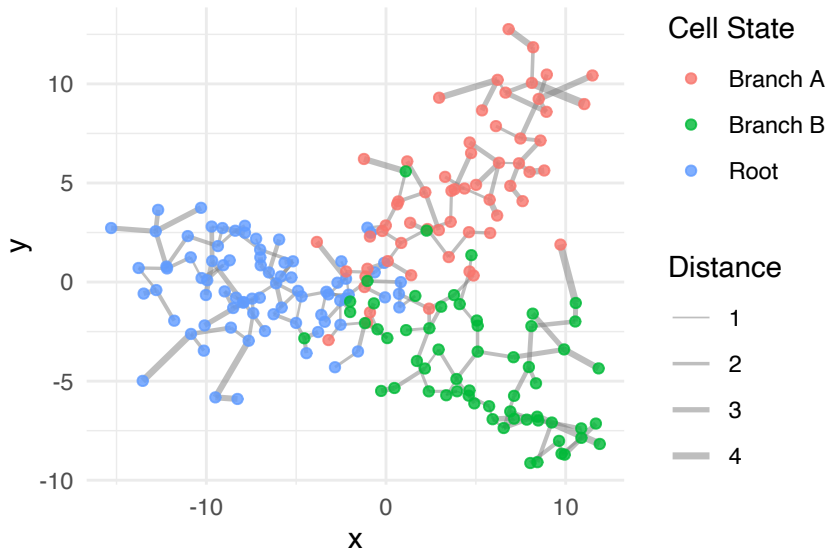


Building Minimum Spanning Tree

Edges: 100

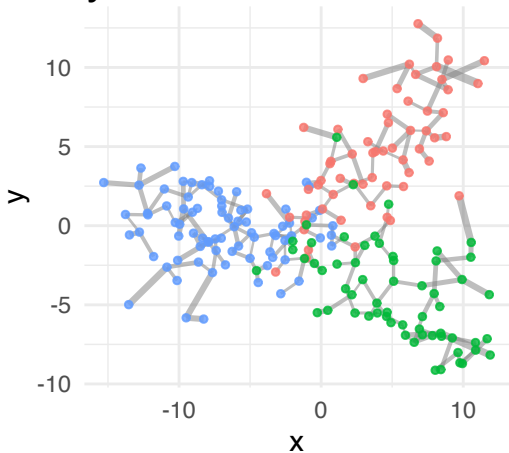


Build minimum spanning tree (MST)

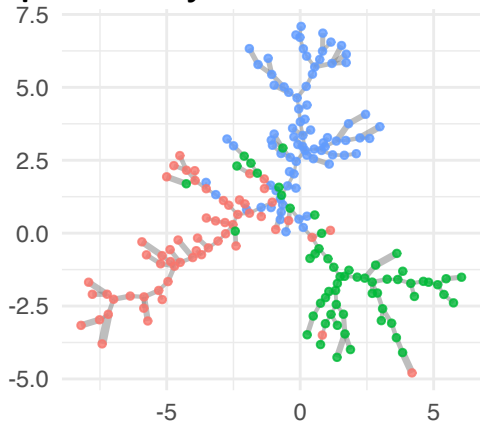


Graph-based 2D layout show the MST more explicitly

PCA layout

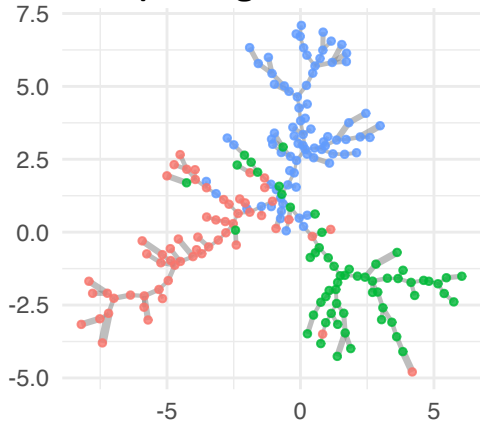


Graph-based layout

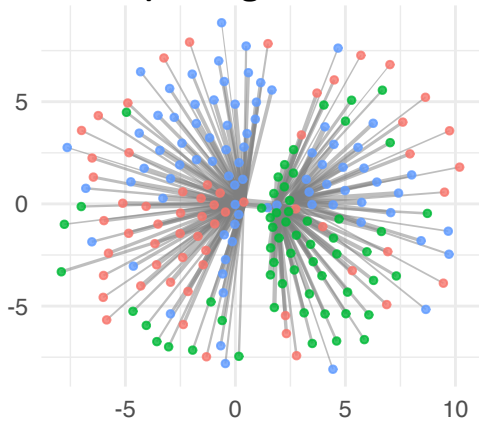


What if we build Maximum Spanning Tree?

Minimum Spanning Tree

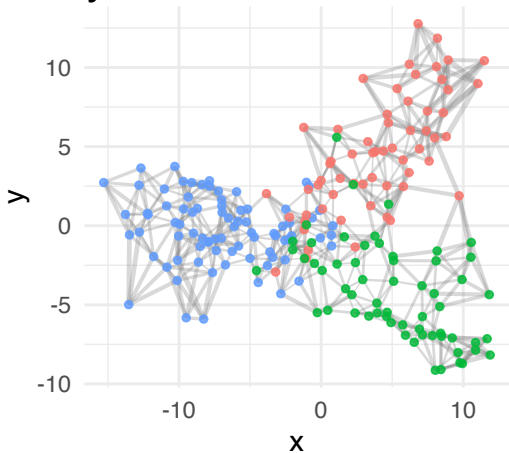


Maximum Spanning Tree

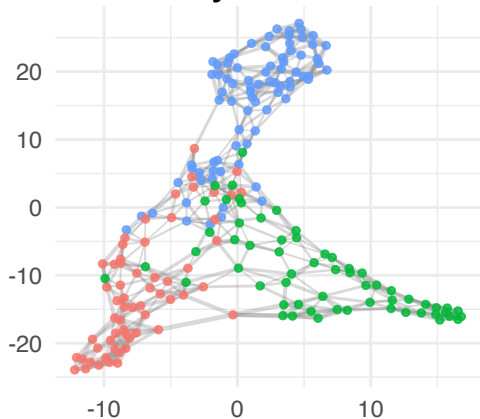


Alternative: k-Nearest Neighbor (kNN) Graph

PCA layout



Force-directed layout



Let's refine our paragraph on the problem definition.

- ① We consider each cell as a point along a differentiation trajectory. We assume that **most genes are affected by the differentiation process**, so cell-to-cell distances are best captured in “principal” component analysis.
- ② We represent the relationships between cells as a graph, **more precisely, a tree structure reflecting their developmental lineage**.
- ③ Since gene expressions are **gradually changed** over the differentiation process, each **branch of the underlying tree structure** represents distinctive cell types.
- ④ We annotate cell type labels based on prior knowledge (more work needed).

More formal definition of the problem

Input

- Gene expression X (m genes $\times$ n cells)
- Assumption: a differentiation process (minimum spanning tree)

Output

- Cell-to-cell MST graph
- Assign each cell vector $\mathbf{x}_j$ to label $c_i \in \{1, \dots, K\}$

More specific problem definition of MST

Input

Distance graph $G = (V, E, D)$ where

- V = vertices (cells)
- E = edges between cells
- D = distance matrix

Output

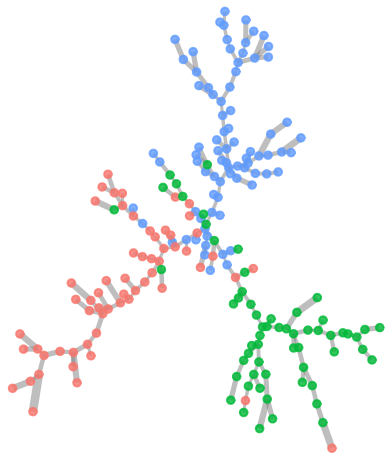
Tree T connecting all cells

Objective

Minimize total edge weight (MST)

$$\min_T \sum_{(i,j) \in E(T)} D_{ij}$$

Optional: more on identifying branches in the tree



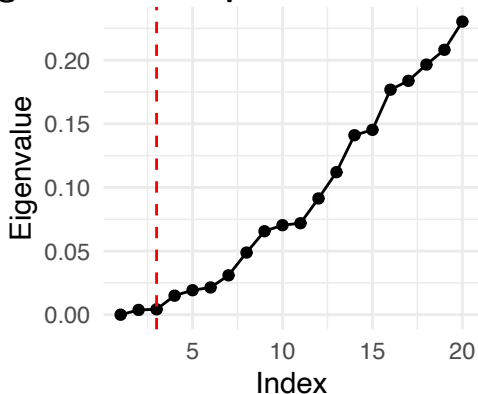
state

- Branch A
- Branch B
- Root

- True “state” is hidden (need to estimate)
- We will revisit how we can do clustering on graphs
- But the following three slides are there for the sake of completeness

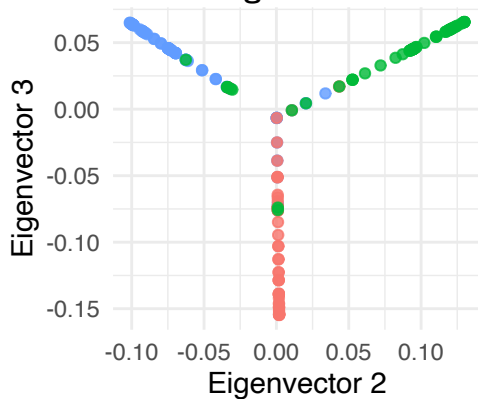
Optional: Spectral clustering on MST

Eigenvalues of Laplacian



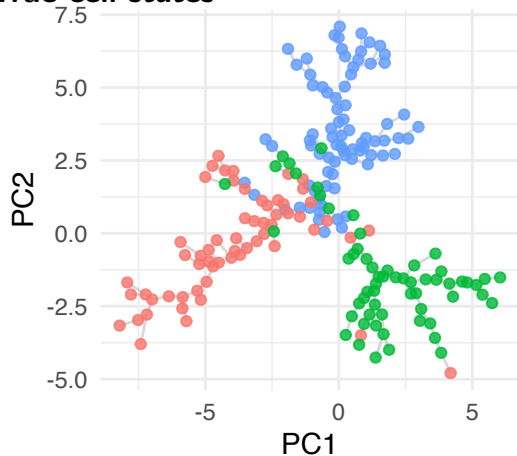
Gap at $k=3$ suggests 3 clusters

Spectral embedding

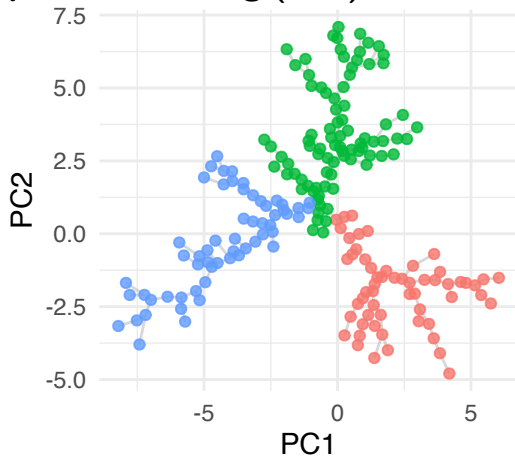


Optional: Spectral cut results

True cell states



Spectral clustering (k=3)



Today's lecture

- 1 Define a problem as an optimization problem
- 2 Use probabilities to describe data generation process

Why Probabilistic (Graphical) Model?

PGM forces us to be explicit about:

- ① **What we observe** (data)
- ② **What we want to learn** (parameters, latent variables)
- ③ **What assumptions we make** (model structure)
- ④ **What we don't know** (uncertainty)

“All models are wrong, but some are useful.” – George Box

Describe your science by stitching together random variables

Represent joint probability distributions using graphs

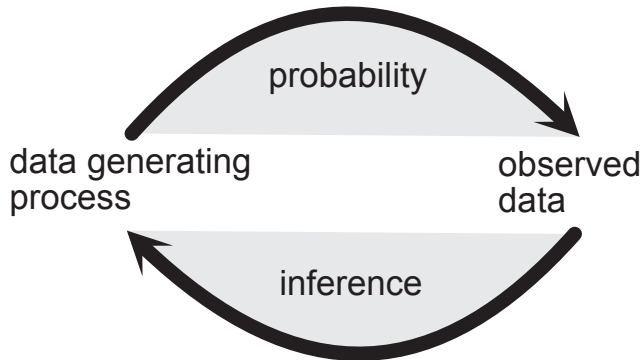
Nodes = Random variables

- Observed (shaded)
- Latent/hidden (open)
- Parameters (dots or small nodes)

Edges = Dependencies

- Arrow $\rightarrow$ = causal/generative direction
- No arrow = correlation/association

lect-02: What is statistics?



Wasserman, *All of Statistics* (2003)

As long as you can write out the probability...

Central question: How did we get our data?

- ① **Write down your data generating model**
- ② Inference: Fit observed data to the model
- ③ Validation: How well? How significant?

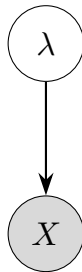
Example: the number of mutations in HIV (lect02)

Let's graphically represent

$$X \mid \lambda \sim \text{Poisson}(\lambda)$$

where

- λ : unknown mutation rate parameter (open)
- X : observed number of mutations (shaded)



Revisit the same question: cell type identification

Data: RNA-seq of **genes** $\times$ **cells**

Let X_{gi} be the expression value of a gene g in a cell i .

$$\mathbf{X} = \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1n} \\ X_{21} & X_{22} & \cdots & X_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ X_{m1} & X_{m2} & \cdots & X_{mn} \end{pmatrix} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$$

where m corresponds to 19k genes.

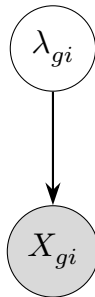
The number of mRNA counts in the same way?

Let's graphically represent

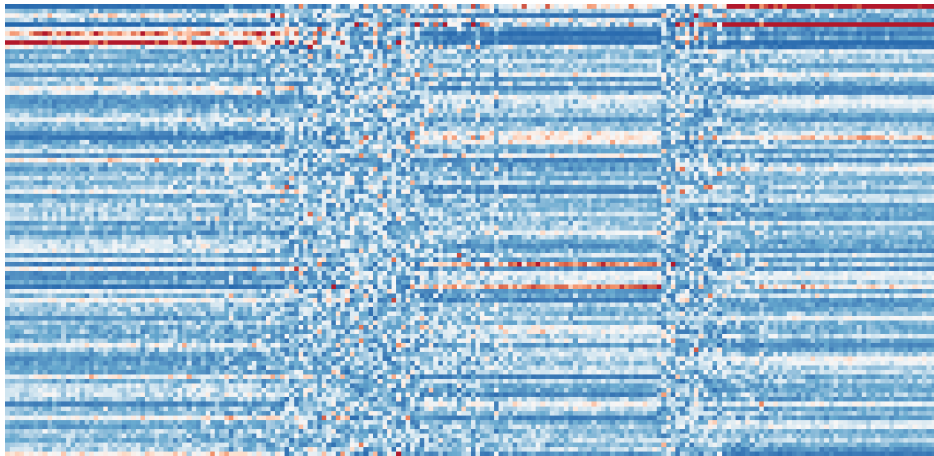
$$X_{gi} \mid \lambda \sim \text{Poisson}(\lambda_{gi})$$

where

- λ_{gi} : unknown transcription rate parameter of a gene g within a cell i (open; latent; parameter)
- X_{gi} : the number of mRNA reads of a gene g in a cell i (shaded; observed; data)



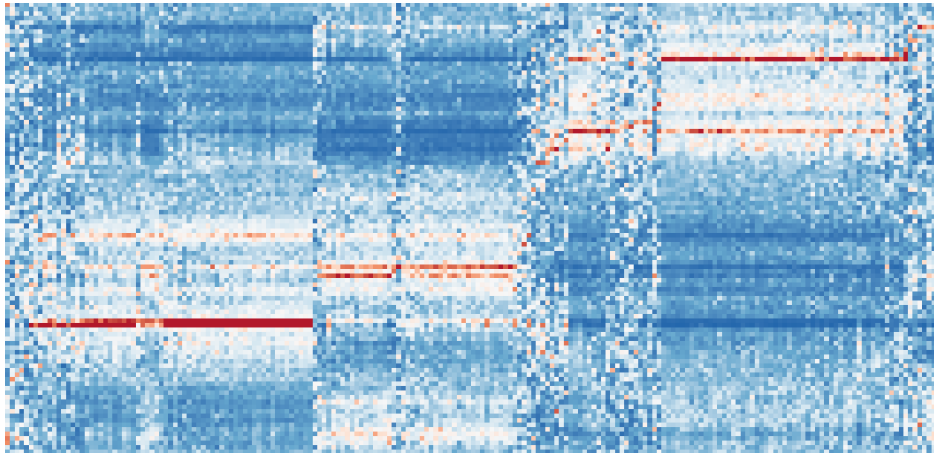
Can we elaborate more on the λ_{gi} (gene $\times$ cell)?



What's wrong with having λ_{gi} for each X_{gi} ?

Let's think of the “generation” of X (or λ)

HINT (just sorted rows and columns):



Write out data generating scheme

Of the K cell types:

$$Z_{ik} = 1 \iff \text{cell } i \text{ in type } k$$

Write out data generating scheme

Of the K cell types:

$$Z_{ik} = 1 \iff \text{cell } i \text{ in type } k$$

For each cell type k , a gene g is expressed with the rate

$$\lambda_{gk}$$

Note: do we have more λ_{gi} 's
or λ_{gk} 's?

Write out data generating scheme

Of the K cell types:

$$Z_{ik} = 1 \iff \text{cell } i \text{ in type } k$$

For each cell type k , a gene g is expressed with the rate

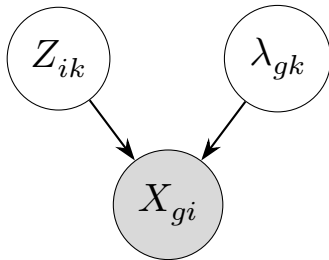
$$\lambda_{gk}$$

Note: do we have more λ_{gi} 's
or λ_{gk} 's?

Given the cell type assignment Z_{ik} and type specific expression rate parameters λ , we have

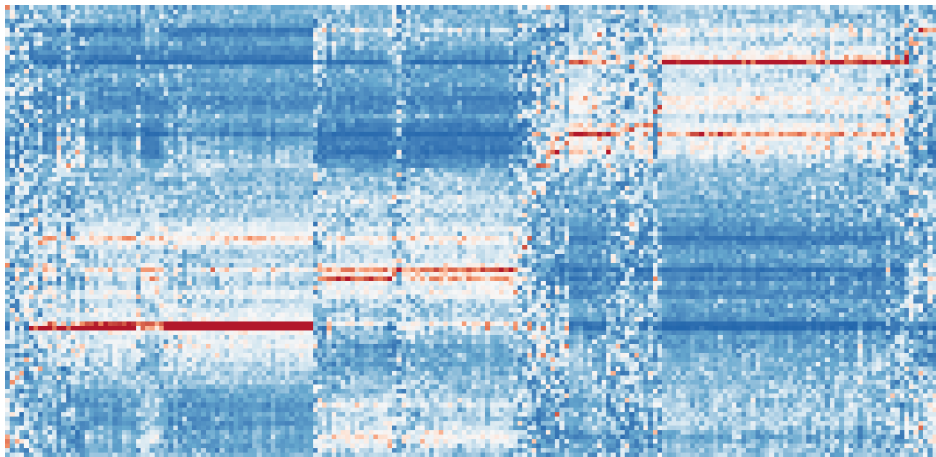
$$X_{gi} | Z_{ik} = 1 \sim \text{Poisson}(\lambda_{gk})$$

Or we could simply draw it out



What's next?

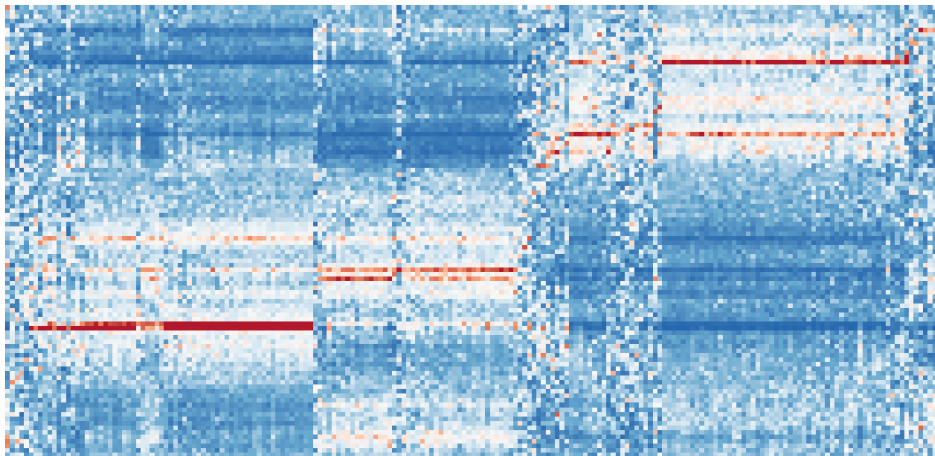
Where do these Z and λ variables correspond to?



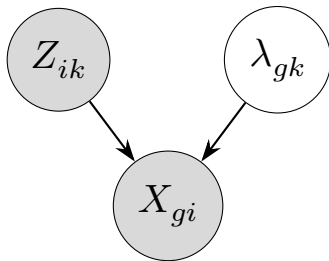
Where do these Z and λ variables correspond to?

$$Z_{ik} \rightarrow \rightarrow \rightarrow$$

$$\lambda_{gk}$$



In many cases we have assignment Z observed/given



What's next (lect04)?

$$\lambda_{g1} = \lambda_{g2} \quad \text{or} \quad \lambda_{g1} \neq \lambda_{g2}$$

As long as we have PGM...

- Estimate the unknown parameter, such as λ
- Check how “well” generated data, such as X , can explain observed X

Summary

Two approaches to formulate computational problems:

① Optimization approach

- Define input, output, metric
- The art of abstraction
- Example: MST for cell trajectories

② Probabilistic modeling

- Write down generative model
- Use graphical models (PGM)
- Example: Poisson model for gene expression

Key takeaway: Write down your problem clearly before coding!